

EDWARD JENNER

English doctor and chemist
1749–1823

Smallpox

Smallpox had been a dreaded disease for thousands of years. Edward's cowpox vaccinations made it possible to prevent it. It was an amazing discovery, given that no one then knew what viruses were or how immune systems worked! Before the vaccinations, three out of every 10 people who caught smallpox died, and survivors were often left with terrible scars.



Cowpox blisters

The hunt for a coronavirus vaccination became headline news in 2020, but the story of this important method of disease prevention began more than 200 years ago. As a doctor in the 18th century, Edward Jenner often saw patients with a terrible disease called smallpox. He noticed a pattern: farm workers who had recovered from a similar, yet milder, disease called cowpox seemed to be protected from smallpox.

Edward tested this theory by infecting a nine-year-old boy with cowpox on purpose. Two months later, he dabbed the child with pus from a smallpox patient, but the boy didn't get sick. Edward called his new treatment "vaccination," after the Latin word for cow: *vacca*. Edward's discovery was very successful and saved millions of lives. Eventually, by 1980, the disease was eradicated—there are no more smallpox cases anywhere in the world.



As a patient's body learned to fight the cowpox virus, it also learned to fight the similar smallpox virus.